

CS 725: Quiz 1

Roll No.: _____

Credit/Audit/Sit-Through: _____

Note: For multi-correct type of questions, one or more option(s) might be correct. You will be awarded full marks if and only if you have selected all correct answers and no wrong answers. There will be no partial marks.

1. Consider $\mathbf{v}_1 = (1, 2, 1)$, $\mathbf{v}_2 = (2, 1, 3)$, $\mathbf{v}_3 = (3, 3, 4)$ in $\mathbb{R}^3$. Which statement is correct?

- (a) The set is linearly independent and spans $\mathbb{R}^3$
- (b) The set is linearly dependent and spans a 2-dimensional subspace.
- (c) The set is linearly dependent and spans a 1-dimensional subspace.
- (d) The set is linearly independent and spans a 2-dimensional subspace.

..1

Correct: (b)

$\mathbf{v}_3 = \mathbf{v}_1 + \mathbf{v}_2 \implies$ linearly dependent

2. Let A be a 4×4 matrix with $\text{rank}(A) = 3$. Which of the following must be true?

- (a) The columns of A are linearly independent
- (b) $\det(A) = 0$
- (c) $Ax = b$ has a unique solution for every $b \in \mathbb{R}^4$
- (d) $Ax = 0$ has infinitely many solutions

..2 (multicorrect)

Correct: (b), (d)

$\text{rank}(A) = 3 < 4$ means $\det(A) = 0$, and $\text{nullity}(A) = 1$ gives infinitely many solutions to $Ax = 0$.

3. Let $A = \begin{bmatrix} 1 & 2 & 0 \\ 3 & -1 & 4 \\ 2 & 5 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 5 & -3 & 2 \\ 1 & 4 & -6 \\ 0 & 7 & -2 \end{bmatrix}$, $C = \begin{bmatrix} -3 & 3 & -2 \\ -1 & -5 & 6 \\ 0 & -7 & 5 \end{bmatrix}$. Compute $AB + AC$.

(a) $\begin{bmatrix} 2 & 4 & 0 \\ -3 & 1 & -4 \\ 6 & 15 & 3 \end{bmatrix}$

(b) $\begin{bmatrix} 2 & -2 & 0 \\ 6 & 1 & 12 \\ 4 & -5 & 3 \end{bmatrix}$

(c) $\begin{bmatrix} 3 & 2 & 0 \\ 3 & -2 & 4 \\ 2 & 5 & 4 \end{bmatrix}$

(d) $\begin{bmatrix} 2 & 2 & 0 \\ 6 & -1 & 12 \\ 4 & 5 & 3 \end{bmatrix}$

..2

Correct: (b)

Use commutative property.

4. Let Q be a real $n \times n$ orthogonal matrix. Which of the following must be true?

- (a) $Q^{-1} = Q^T$
- (b) $\|Qx\| = \|x\|$ for every $x \in \mathbb{R}^n$
- (c) $\det(Q) = 1$
- (d) Every eigenvalue of Q is real

..2 (multicorrect)

Correct: (a), (b)

5. Find the inverse of $A = \begin{bmatrix} 2 & 1 \\ 5 & 3 \end{bmatrix}$.

- (a) $\begin{bmatrix} 3 & -1 \\ -5 & 2 \end{bmatrix}$
- (b) $\begin{bmatrix} 3 & 1 \\ 5 & 2 \end{bmatrix}$
- (c) $\begin{bmatrix} 1/2 & 1 \\ 1/5 & 1/3 \end{bmatrix}$
- (d) $\begin{bmatrix} 2 & 5 \\ 1 & 3 \end{bmatrix}$

..1

Correct: (a)

6. The eigenvalues of $A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$ are p and q . What is the value of $|p - q|$? _____ ..1

Correct: 3

$$\det(A - \lambda I) = \lambda^2 - 7\lambda + 10 = 0 \implies \lambda = 5, 2$$

7. For the matrix $A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$, what the eigenvector corresponding to the smaller eigenvalue?

- (a) (1, 1)
- (b) (1, -2)
- (c) (2, 1)
- (d) (1, 2)

..1

Correct: (b)

$$(A - 2I) \cdot (x_1, x_2) = \begin{bmatrix} 2 & 1 \\ 2 & 1 \end{bmatrix} \cdot (x_1, x_2) = 0 \implies 2x_1 + x_2 = 0 \implies \text{The eigenvector is } (1, -2)$$

8. Let A and B be 4×4 matrices with $\det(A) = 3$ and $\det(B) = -2$. $\det(2A^\top B^{-1}) = \underline{\hspace{2cm}}$.
..1

Correct: -24

$$\det(2A^\top B^{-1}) = \det(2A^\top) \times \det(B^{-1}) = 2^4 \times \det(A^\top) \times \det(B^{-1}) = 2^4 \times 3 \times (-\tfrac{1}{2}) = -24$$

Note: Full marks even if you have answered 24 (thanks to the SAFE bug :)).

9. Which of the following symmetric matrices is positive definite?

(a) $\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$

(b) $\begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$

(c) $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

(d) $\begin{bmatrix} -2 & 0 \\ 0 & -3 \end{bmatrix}$

..1

Correct: (a)

Check if all eigenvalues are positive.

10. Let A be a real symmetric positive definite $n \times n$ matrix. Which of the following must be true?

(a) All diagonal entries of A are positive

(b) Every entry of A is positive

(c) A must have n distinct eigenvalues

(d) A is invertible and A^{-1} is also positive definite

..2 (multicorrect)

Correct: (a), (d)

Total: 14